# Natalyx MVP Fixes

Work Package 08: Payments, invoices, credits, and refunds

L. Rhodes (plan owner) - A. Zhao (technical review requested)

2026-07-10

# Contents

**1 Work Package 08: Payments, invoices, credits, and refunds** **2**

1.1 Purpose and outcome . . . . . 2

1.2 Verified current-state evidence . . . . . 3

1.3 Architecture and contracts . . . . . 3

1.4 Explicit PR sequence . . . . . 9

1.5 Cross-repository coordination . . . . . 17

1.6 Test and acceptance matrix . . . . . 18

1.7 Risks and decisions . . . . . 19

1.8 Existing-work salvage . . . . . 21

1.9 Ownership recommendation . . . . . 22

1.10 Definition of done . . . . . 22

# 1 Work Package 08: Payments, invoices, credits, and refunds

Review draft only. Ownership and sequencing remain recommendations until Luke and Alan approve the plan together.

| Plan | Value |
|---|---|
| Work package | WP08 |
| Plan file | `WP08 - Payments, invoices, credits, and refunds.md` |
| Proposed PRs | P08-01, P08-02, P08-03, P08-04 |
| Inspection snapshot | 2026-07-10 20:12 America/New_York |

## 1.1 Purpose and outcome

**Problem.** No append-only commercial ledger or processor-neutral payment boundary currently supports the $8,900 JMF, $750 enrollment, Node 0 credit, milestones, deterministic refunds, or failure recovery.

**Why this package exists.** It creates a reviewable boundary so the programme can progress without mixing product, data, security, infrastructure and rollout concerns in one oversized change.

**Completed outcome.** Immutable commercial ledgers, idempotent provider inbox, milestone engine, reproducible credits/refunds, reconciliation, and safe non-critical pause behavior.

**Explicit non-goals.**

- No custody of pass-through funds

- No financing assumption
- No pregnancy-loss automation before policy
- No accounting-general-ledger replacement

## 1.2 Verified current-state evidence

Classification is explicit: **verified** is observed code/PR/live behavior; **verified limitation** records an evidence gap; **design authority** is an approved planning source; **design decision** is proposed future behavior. None is silently treated as another.

| ID | Class | Repository / ref | File, construct or location | Finding |
|---|---|---|---|---|
| E-025 | design authority | business model / Natalyx_Payment_and_Business_Model_BYOS_MVP.docx v1.0 | Sections 3-12 and implementation tables | Node 0 Free must complete the match; Complete is optional/creditable; $8,900 JMF and $750 enrollment anchor the configurable model; promotion follows explicit commitment and funding, not willingness or redirect. |

## 1.3 Architecture and contracts

- Payments, credits, refunds, adjustments, invoices, and commercial events are append-only; mutable balances are derived projections.
- Processor events enter an idempotent inbox and become internal commands/events only after signature and amount/currency checks.
- Billing follows durable case evidence and never acts as first notification of clinical events.

| Contract area | Package rule |
|---|---|
| Ownership boundary | `app`; repository-specific PRs remain separate. |
| Entities/state | `journey_commitment`, `payment_obligation + append-only entries`, `workspace_entitlement` |
| Authorization | Authenticated subject/service plus entity membership, role, purpose, consent and expected revision; query parameters, links and UI state are never authority. |
| API/service | Processor webhook and internal payment commands. Internal evidence-to-invoice commands and explainable refund quote. |
| Storage | Append-only financial ledgers and unique idempotency constraints. Processor-event inbox/outbox and mapping records. |
| Audit | Use the common transition envelope and append-only source/event identifiers; redact tokens, raw documents, prompts and sensitive payloads. |
| Failure | Fail closed for authorization, provenance, money, document and PHI boundaries; expose a bounded actionable error and preserve retry/repair evidence. |

| Contract area | Package rule |
|---|---|
| Observability | Correlation, policy/schema/config version, transition and dependency result metrics without PHI/PII payload logging. |
| Compatibility | Additive/dual-compatible changes, explicit version pins, old/new revision window and named forward repair. |
| Privacy/PHI | Synthetic only through A2; approved PII only at B1; PHI remains blocked until formal B2. |

### 1.3.1 Applicable state machines

### 1.3.2 Entitlement (`workspace_entitlement`)

Owner: **natalyx-app commercial domain**. States: `free`, `complete_pending`, `complete_active`, `suspended`, `revoked`.

**Transition: `none` -> `free`**

- **Actor:** system
- **Evidence:** active Node 0 workspace
- **Idempotency key:** `entitlement:{workspace}:free`
- **Invalid transitions:** No workspace
- **Audit event:** `entitlement_granted`
- **Retry:** Return current
- **Recovery:** Repair projection from event

**Transition: `free` -> `complete_pending`**

- **Actor:** IP/commercial service
- **Evidence:** accepted Complete offer/config
- **Idempotency key:** `entitlement:{workspace}:{commitment}`
- **Invalid transitions:** Unknown price/config/terms
- **Audit event:** `complete_requested`
- **Retry:** Return pending
- **Recovery:** Cancel/expire commitment

**Transition: `complete_pending` -> `complete_active`**

- **Actor:** payment domain
- **Evidence:** settled Complete ledger entry
- **Idempotency key:** `entitlement:{payment}:activate`
- **Invalid transitions:** Amount/currency/config mismatch
- **Audit event:** `complete_activated`
- **Retry:** Replay no-op
- **Recovery:** Reconcile and retry from inbox

**Transition: `complete_active` -> `suspended`**

- **Actor:** authorized commercial policy
- **Evidence:** refund/reversal or approved pause reason
- **Idempotency key:** `entitlement:{id}:suspend:{event}`
- **Invalid transitions:** Cannot remove free/safety capability
- **Audit event:** `entitlement_suspended`
- **Retry:** No-op if same event
- **Recovery:** Reinstate or revoke through new event

**Transition:** `complete_active|suspended -> revoked`

- **Actor:** authorized operator/policy
- **Evidence:** fraud/refund/terms revocation evidence
- **Idempotency key:** `entitlement:{id}:revoke:{event}`
- **Invalid transitions:** No silent case cancellation
- **Audit event:** `entitlement_revoked`
- **Retry:** Idempotent
- **Recovery:** Manual reviewed regrant creates new entitlement event

#### 1.3.3 Commercial commitment (`journey_commitment`)

Owner: **natalyx-app commercial domain**. States: `draft`, `accepted`, `funding_pending`, `funded`, `cancelled`, `expired`.

**Transition:** `none -> draft`

- **Actor:** IP
- **Evidence:** active workspace + displayed config/terms
- **Idempotency key:** `commitment:{workspace}:{config}`
- **Invalid transitions:** No active config/workspace
- **Audit event:** `commitment_drafted`
- **Retry:** Return existing draft
- **Recovery:** Create new draft after expiry/version change

**Transition:** `draft -> accepted`

- **Actor:** IP
- **Evidence:** terms/privacy consent + payment path
- **Idempotency key:** `commitment:{id}:accept:{terms_hash}`
- **Invalid transitions:** Stale/missing acceptance
- **Audit event:** `commitment_accepted`
- **Retry:** Return accepted
- **Recovery:** New acceptance revision if version changes

**Transition:** `accepted -> funding_pending`

- **Actor:** commercial service
- **Evidence:** enrollment invoice/payment intent created

- **Idempotency key:** `commitment:{id}:funding`
- **Invalid transitions:** No invoice or invalid amount
- **Audit event:** `funding_requested`
- **Retry:** Resume same invoice
- **Recovery:** Cancel/reissue under audited adjustment

**Transition: `funding_pending` -> `funded`**

- **Actor:** payment domain
- **Evidence:** net settled enrollment funding meets versioned requirement
- **Idempotency key:** `enrollment-funding:{id}:v1`
- **Invalid transitions:** Redirect/pending/failed/reversed evidence
- **Audit event:** `commercial.enrollment_funding_confirmed.v1`
- **Retry:** Exactly-once event/outbox replay
- **Recovery:** Reconcile inbox/ledger; no browser repair

**Transition: `draft|accepted|funding_pending` -> `cancelled`**

- **Actor:** IP/authorized operator
- **Evidence:** explicit cancellation and refund policy
- **Idempotency key:** `commitment:{id}:cancel:{request}`
- **Invalid transitions:** Already funded/promoted uses case cancellation rail
- **Audit event:** `commitment_cancelled`
- **Retry:** Idempotent
- **Recovery:** New commitment required

**Transition: `draft|accepted` -> `expired`**

- **Actor:** system
- **Evidence:** offer/terms TTL or config deactivation
- **Idempotency key:** `commitment:{id}:expire`
- **Invalid transitions:** No expiry after funding
- **Audit event:** `commitment_expired`
- **Retry:** Idempotent
- **Recovery:** Draft new version

#### 1.3.4 Payment (`payment_obligation + append-only entries`)

Owner: **natalyx-app payment domain**. States: `created`, `processing`, `settled`, `failed`, `partially_refunded`, `refunded`, `reversed`.

**Transition: `none` -> `created`**

- **Actor:** billing policy
- **Evidence:** valid invoice/obligation
- **Idempotency key:** `payment:{invoice}:create`
- **Invalid transitions:** Duplicate or invalid invoice

- **Audit event:** `payment_created`
- **Retry:** Return current intent
- **Recovery:** Void/reissue through adjustment

**Transition:** `created|failed -> processing`

- **Actor:** processor adapter
- **Evidence:** signed provider acknowledgement
- **Idempotency key:** `processor:{provider_event}`
- **Invalid transitions:** Amount/currency/customer mismatch
- **Audit event:** `payment_processing`
- **Retry:** Inbox replay no-op
- **Recovery:** Retry same intent or audited replacement

**Transition:** `processing|created -> settled`

- **Actor:** payment domain
- **Evidence:** verified settlement event
- **Idempotency key:** `processor:{provider_event}`
- **Invalid transitions:** Browser redirect or unverified event
- **Audit event:** `payment_settled`
- **Retry:** Exactly-once append
- **Recovery:** Reconcile provider and append repair event

**Transition:** `created|processing -> failed`

- **Actor:** payment domain
- **Evidence:** verified failure/timeout policy
- **Idempotency key:** `processor:{provider_event}`
- **Invalid transitions:** Cannot overwrite settled
- **Audit event:** `payment_failed`
- **Retry:** Replay no-op
- **Recovery:** Retry/new attempt linked to obligation

**Transition:** `settled|partially_refunded -> partially_refunded|refunded`

- **Actor:** authorized refund service
- **Evidence:** approved refund calculation + processor confirmation
- **Idempotency key:** `refund:{refund_request}`
- **Invalid transitions:** Exceeds settled less prior refunds
- **Audit event:** `refund_settled`
- **Retry:** Replay no-op
- **Recovery:** Reconcile/adjust; never mutate original

**Transition:** `settled -> reversed`

- **Actor:** payment domain

- **Evidence:** verified chargeback/reversal
- **Idempotency key:** `processor:{provider_event}`
- **Invalid transitions:** Unsupported/manual assertion
- **Audit event:** `payment_reversed`
- **Retry:** Replay no-op
- **Recovery:** Open discrepancy and apply policy; case is not deleted

#### 1.3.5 Append-only commercial contract

Authoritative ledgers:

- `commercial_events`
- `processor_events`
- `invoice_events`
- `payment_ledger_entries`
- `credit_ledger_entries`
- `refund_ledger_entries`
- `adjustment_ledger_entries`
- `performance_obligation_events`
- `promotion_events`

Derived projections:

- payment plan state
- invoice status
- settled cash
- available credit
- recognized/deferred amount
- refundable amount
- delinquency/pause status

Rules:

- Every record carries currency in integer minor units, commercial configuration version, source/idempotency/correlation identifiers and immutable timestamps.
- Mutable balances/statuses are projections only; reversals, chargebacks, credits, refunds and adjustments append entries.
- Processor events are signature-verified, stored once and normalized before domain use.
- Node 0 Complete credit cannot create a negative JMF obligation and stays distinct from cash/refund entries.
- Pass-through clinic/legal/psych/background/escrow/insurance/GC funds never enter Natalyx revenue ledgers.
- Clinical-linked invoices require durable evidence plus dignity-of-timing delay and can never be the first clinical notification.
- Refundable cash = max(0, settled JMF cash - earned JMF - prior cash refunds - chargebacks/reversals), with policy hooks for unresolved cases.

## 1.4 Explicit PR sequence

| Order | ID | Title | Repository | Dependencies | Gate | Size | Readiness |
|---|---|---|---|---|---|---|---|
| 1 | P08-01 | Add append-only commercial and money ledgers | app | P02-03, P07-01 | A2 | Exceptional | Dependencies unmet |
| 2 | P08-02 | Add processor-neutral payment command and webhook inbox | app | P08-01, P07-02 | A2 | Exceptional | Blocked by decision |
| 3 | P08-03 | Implement milestone invoicing, credits, and deterministic refunds | app | P08-02, P07-04, P09-04 | A2 | Exceptional | Dependencies unmet |
| 4 | P08-04 | Add payment reconciliation and safe service-pause operations | app | P08-03 | A2 | Large | Dependencies unmet |

### 1.4.1 P08-01 - Add append-only commercial and money ledgers

| Field | Plan |
|---|---|
| Repository / base | `app` (`main`) |
| Gate / size / readiness | A2 / Exceptional / Dependencies unmet |
| Dependencies | P02-03, P07-01 |
| Blocking decisions/checkpoints | D-COM-01, D-COM-02, D-COM-03 |
| Source / salvage | Payment/business-model DOCX |

**1.4.1.1 Objective** Create immutable commercial-event, invoice, payment, credit, refund, and adjustment entries with deterministic projections and config-version references.

#### 1.4.1.2 Scope

- Ledger invariants and currency precision
- Derived balances/status projections
- Audit/source/idempotency references

#### 1.4.1.3 Explicit non-goals

- No processor
- No mutable balance authority
- No pass-through custody

#### 1.4.1.4 Expected files or modules

- `migrations/versions/*_commercial_ledgers.py`
- `app/storage/ledgers.py`
- `app/domain/money.py`
- `tests/python/test_ledgers.py`

#### 1.4.1.5 Change map

| Area | Planned change |
|---|---|
| Schema/model | Append-only financial ledgers and unique idempotency constraints. |
| API/routes | No public API/route contract change. |
| UI/product surface | No visible participant/operator UI change. |
| Infrastructure/workflow | No cloud/infrastructure resource or workflow change. |
| Migration/backfill | Apply through the named versioned migration; no ad-hoc startup DDL. Backfill is none unless explicitly listed in scope. |
| Security/privacy | Least-privilege write paths and redacted logs. |
| Architecture/docs/config | Document the contract, version pins and safe partial-merge behavior. |

**1.4.1.6 Tests and validation** Objective tests:

- Double-entry-like conservation checks
- duplicate key
- credit floor
- refund totals
- concurrent append

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate **A2** on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

**1.4.1.7 Diagnostics and observability** Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

**1.4.1.8 Rollout** Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

**1.4.1.9 Rollback or forward repair** Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

**1.4.1.10 Dependencies and parallel work** Hard dependencies: P02-03, P07-01. Parallel opportunities: P08-03, P08-04

#### 1.4.1.11 Objective acceptance criteria

- ☐ Create immutable commercial-event, invoice, payment, credit, refund, and adjustment entries with deterministic projections and config-version references.
- ☐ Every listed objective test passes locally and at the required gate where applicable.
- ☐ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- ☐ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

#### 1.4.1.12 Ownership and review

| Role | Recommendation |
|---|---|
| DRI | Luke |
| Implementer | Luke (proposed; pending joint approval) |
| Reviewer | Alan |
| Final approver | Joint |
| Required expertise | payments, financial ledgers, PostgreSQL |
| Higher-capability review | Mandatory before merge |

### 1.4.2 P08-02 - Add processor-neutral payment command and webhook inbox

| Field | Plan |
|---|---|
| Repository / base | `app` (`main`) |
| Gate / size / readiness | A2 / Exceptional / Blocked by decision |
| Dependencies | P08-01, P07-02 |
| Blocking decisions/checkpoints | D-PAY-01, D-COM-05 |
| Source / salvage | Business-model processor-neutral requirement |

**1.4.2.1 Objective** Define provider ports and a signed, idempotent webhook inbox that emits internal settlement/failure/refund events, including commercial.enrollment_funding_confirmed.v1 when domain conditions hold.

#### 1.4.2.2 Scope

- Provider-neutral intents and event mapping
- Signature/replay/order handling
- Settlement predicate and outbox

#### 1.4.2.3 Explicit non-goals

- No browser redirect settlement
- No financing implementation unless separately approved

#### 1.4.2.4 Expected files or modules

- `app/payments/ports.py`
- `app/payments/webhook_inbox.py`
- `app/routes/payment_webhooks.py`
- `tests/python/test_payment_webhooks.py`

#### 1.4.2.5 Change map

| Area | Planned change |
|---|---|
| Schema/model | Processor-event inbox/outbox and mapping records. |
| API/routes | Processor webhook and internal payment commands. |
| UI/product surface | No visible participant/operator UI change. |
| Infrastructure/workflow | Secrets/endpoints deferred to guarded environment configuration. |
| Migration/backfill | No migration or backfill. |
| Security/privacy | Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases. |
| Architecture/docs/config | Document the contract, version pins and safe partial-merge behavior. |

**1.4.2.6 Tests and validation** Objective tests:

- Signature failure
- replay/out-of-order
- amount/currency mismatch
- timeout/retry
- settlement event exactly once

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate **A2** on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

**1.4.2.7 Diagnostics and observability** Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

**1.4.2.8 Rollout** Ship expand-only and dark by default; run the one-off migrator, verify readiness/old-revision compatibility, then enable writers/readers in a later compatible step.

**1.4.2.9 Rollback or forward repair** Prefer application rollback within the expand/contract window and a new forward-repair migration. An irreversible schema reversal requires its own approval and backup evidence.

**1.4.2.10 Dependencies and parallel work** Hard dependencies: P08-01, P07-02. Parallel opportunities: P08-04

#### 1.4.2.11 Objective acceptance criteria

- ☐ Define provider ports and a signed, idempotent webhook inbox that emits internal settlement/failure/refund events, including commercial.enrollment_funding_confirmed.v1 when domain conditions hold.
- ☐ Every listed objective test passes locally and at the required gate where applicable.
- ☐ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- ☐ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

#### 1.4.2.12 Ownership and review

| Role | Recommendation |
|---|---|
| DRI | Luke |
| Implementer | Luke (proposed; pending joint approval) |
| Reviewer | Alan |
| Final approver | Joint |
| Required expertise | payment security, webhooks, event systems |
| Higher-capability review | Mandatory before merge |

### 1.4.3 P08-03 - Implement milestone invoicing, credits, and deterministic refunds

| Field | Plan |
|---|---|
| Repository / base | `app` (`main`) |
| Gate / size / readiness | A2 / Exceptional / Dependencies unmet |
| Dependencies | P08-02, P07-04, P09-04 |
| Blocking decisions/checkpoints | D-COM-04, D-COM-06, D-COM-07 |
| Source / salvage | Payment/business-model milestone/refund tables |

**1.4.3.1 Objective** Turn explicit evidence-backed case transitions into one invoice each, apply Node 0 credit, and calculate reproducible earned/uncharged/refunded amounts from immutable state.

#### 1.4.3.2 Scope

- Enrollment/activation/clearance/legal-escrow/pregnancy/birth milestones
- Credit allocation
- Non-clearance and cancellation tables
- Dignity delay for clinical-linked invoices

#### 1.4.3.3 Explicit non-goals

- No unresolved pregnancy-loss automatic disposition
- No promise of reimbursement

#### 1.4.3.4 Expected files or modules

- `app/domain/billing_policy.py`
- `app/domain/refunds.py`
- `app/event_handlers/case_billing.py`
- `tests/python/test_billing_policy.py`

#### 1.4.3.5 Change map

| Area | Planned change |
|---|---|
| Schema/model | None in this PR; any later schema change stays in its named migration PR. |
| API/routes | Internal evidence-to-invoice commands and explainable refund quote. |
| UI/product surface | No visible participant/operator UI change. |
| Infrastructure/workflow | No cloud/infrastructure resource or workflow change. |
| Migration/backfill | No migration or backfill. |
| Security/privacy | Only authorized domain events trigger billing. |
| Architecture/docs/config | Document the contract, version pins and safe partial-merge behavior. |

#### 1.4.3.6 Tests and validation

Objective tests:

- All milestone/refund rows
- duplicate/stale transition
- Node 0 credit
- pay-in-full/deposit math
- clinical notification separation

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate **A2** on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

**1.4.3.7 Diagnostics and observability** Emit correlation/idempotency/policy/version identifiers, transition/result counters and bounded failure reasons; never log tokens, processor payloads, raw documents, prompts, PHI or unnecessary PII.

**1.4.3.8 Rollout** Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

**1.4.3.9 Rollback or forward repair** Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

**1.4.3.10 Dependencies and parallel work** Hard dependencies: P08-02, P07-04, P09-04. Parallel opportunities: P08-01

#### 1.4.3.11 Objective acceptance criteria

- ☐ Turn explicit evidence-backed case transitions into one invoice each, apply Node 0 credit, and calculate reproducible earned/uncharged/refunded amounts from immutable state.
- ☐ Every listed objective test passes locally and at the required gate where applicable.
- ☐ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- ☐ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

#### 1.4.3.12 Ownership and review

| Role | Recommendation |
|---|---|
| DRI | Luke |
| Implementer | Luke (proposed; pending joint approval) |
| Reviewer | Alan |
| Final approver | Joint |
| Required expertise | payments, commercial policy, case state |
| Higher-capability review | Mandatory before merge |

### 1.4.4 P08-04 - Add payment reconciliation and safe service-pause operations

| Field | Plan |
|---|---|
| Repository / base | `app` (`main`) |
| Gate / size / readiness | A2 / Large / Dependencies unmet |
| Dependencies | P08-03 |
| Blocking decisions/checkpoints | D-COM-08 |
| Source / salvage | Business-model failure handling |

#### 1.4.4.1 Objective

Expose operator-safe reconciliation, discrepancy, receipt, retry, and non-critical pause controls without blocking safety information or altering clinical truth.

#### 1.4.4.2 Scope

- Ledger-vs-processor reconciliation
- Itemized receipt data
- Grace/reminder/soft-pause policy
- Audited manual adjustments

#### 1.4.4.3 Explicit non-goals

- No organization-wide ambient access
- No critical-service lockout

#### 1.4.4.4 Expected files or modules

- `app/payments/reconciliation.py`
- `app/routes/operator_billing.py`
- `app/templates/internal/billing/`
- `tests/python/test_reconciliation.py`

#### 1.4.4.5 Change map

| Area | Planned change |
|---|---|
| Schema/model | None in this PR; any later schema change stays in its named migration PR. |
| API/routes | No public API/route contract change. |
| UI/product surface | Internal role-gated discrepancy and adjustment surface. |
| Infrastructure/workflow | No cloud/infrastructure resource or workflow change. |
| Migration/backfill | No migration or backfill. |
| Security/privacy | Preserve existing fail-closed boundaries; validate the PR-specific privacy/security cases. |
| Architecture/docs/config | Document the contract, version pins and safe partial-merge behavior. |

#### 1.4.4.6 Tests and validation

Objective tests:

- Role denial
- mismatch workflow
- manual adjustment audit
- critical content remains accessible
- provider outage

Local validation:

- `python -m unittest discover -s tests/python`
- Run the focused test module(s) named above against SQLite and PostgreSQL where applicable.

Cloud validation: run only through Gate **A2** on a known revision, using synthetic data until B1 and no PHI until B2. A merge or green deployment is not gate acceptance.

**1.4.4.7 Diagnostics and observability** Metrics/alerts for settlement, refund, mismatch and reserve inputs.

**1.4.4.8 Rollout** Deploy behind the narrowest practical feature/config gate, verify the named synthetic flow and diagnostics on a known revision, then widen only at the associated programme gate.

**1.4.4.9 Rollback or forward repair** Disable the feature or return to the prior compatible image; retain append-only audit/ledger history and repair with a new event/revision rather than destructive mutation.

**1.4.4.10 Dependencies and parallel work** Hard dependencies: P08-03. Parallel opportunities: P08-01, P08-02

#### 1.4.4.11 Objective acceptance criteria

- ☐ Expose operator-safe reconciliation, discrepancy, receipt, retry, and non-critical pause controls without blocking safety information or altering clinical truth.
- ☐ Every listed objective test passes locally and at the required gate where applicable.
- ☐ Every explicit non-goal remains out of the diff and no protected worktree/branch is modified during extraction.
- ☐ Rollout/repair evidence is attached before merge; unresolved decisions remain blocked rather than encoded as guesses.

#### 1.4.4.12 Ownership and review

| Role | Recommendation |
|---|---|
| DRI | Luke |
| Implementer | Luke (proposed; pending joint approval) |
| Reviewer | Alan |
| Final approver | Joint |
| Required expertise | payment operations, authorization, observability |
| Higher-capability review | Mandatory before merge |

## 1.5 Cross-repository coordination

| Repository | Authority after merge |
|---|---|
| `app` | routing, validation, authorization, participant/case/workflow truth, PHI policy and commercial domain |

Merge order inside this package: `P08-01` → `P08-02` → `P08-03` → `P08-04`.

Handoff rule: producer contract/version must merge and be pinned before its consumer enables the feature. During a partial merge, consumers fail closed or retain the last approved version; they do not fabricate data or silently broaden access.

Compatibility window: keep prior readers/writers and rollback image usable through the named gate; contract/migration PRs document the exact removal point.

## 1.6 Test and acceptance matrix

| Test dimension | Method | Objective result |
|---|---|---|
| Happy path | Webhook replay/order; Ledger invariants; Milestone idempotency; Refund table; Node 0 credit floor; Reconciliation mismatch | Expected state/event/UI and named evidence are produced once. |
| Authorization failure | Wrong/no subject, role, membership, purpose or consent | 403/404 as appropriate; no write or data leak; audit safe failure. |
| Role differences | Separate IP, GC, operator, provider/service identities | Only the approved lens/projection/actions appear. |
| Retries/duplicates | Repeat same canonical request/event | Original result returned; no duplicate membership/case/invoice/document/event. |
| Concurrency | Two writers/claimants/migrators/workers | One valid transition; loser receives deterministic stale/conflict response. |
| Stale state | Old revision/config/terms/token/artifact | Fail closed or approved compatibility behavior; never silent coercion. |
| Malformed input | Boundary, encoding, size, enum, redirect and identifier corpus | Validation before side effects; safe error. |
| Unavailable dependency | Database, intelligence, processor, S3/scanner or auth unavailable | Explicit degraded/blocked behavior; no hallucinated success. |
| Partial failure | Crash between external effect, DB transaction and outbox | Retry/forward repair converges from durable evidence. |
| Logout/login/resume | Session expires/restarts mid-flow | Return to authorized current state; no query/link authority. |
| Mobile/accessibility | 320/375/768 widths, keyboard, focus, reduced motion | Core task completes with truthful state and no clipped controls. |
| Synthetic cloud | Known dev revision and synthetic identities/data | Gate artifact records SHA, dependency status, audit and cleanup. |

| Test dimension | Method | Objective result |
|---|---|---|
| Audit/observability | Inspect events/metrics/logs | Versions, actor/authority, correlation and reason present; no sensitive payload. |

## 1.7 Risks and decisions

| Type | ID / risk | Owner | Affected PRs / latest point | Treatment |
|---|---|---|---|---|
| Blocking/product/compliance decision | D-COM-01 - Set the flat one-time price and tax treatment; it must remain fully creditable against the later JMF. | Luke | P06-01, P07-01, P08-01; Before activation of a Complete configuration | D-COM-01 - Keep no active production value until explicitly approved; test with synthetic configuration. |
| Blocking/product/compliance decision | D-COM-02 - Ratify $8,900 JMF, $7,830 pay-in-full, $750 deposit and 20/30/20/20/10 post-deposit milestones as configuration v1 with effective date/currency. | Luke | P07-01, P08-01, P08-03; Before P07-01 activation data | D-COM-02 - Use the document anchors but require explicit version activation rather than code literals. |
| Blocking/product/compliance decision | D-COM-03 - Approve the actual paid-service terms, privacy/consent disclosure and refundable-to-earned wording. | Luke | P07-01, P07-02, P08-03; Before P07-02 user acceptance | D-COM-03 - External legal/privacy/finance review before any real acceptance. |

| Type | ID / risk | Owner | Affected PRs / latest point | Treatment |
|---|---|---|---|---|
| Blocking/product/compliance decision | D-COM-04 - Define evidence for enrollment funding confirmed, promotion authorization, and for onboarding/compatibility complete enough to move case active and earn/issue activation billing. | Luke | P07-02, P07-03, P07-04, P08-03, P09-04; Before P07-02 | D-COM-04 - Funding confirmation is durable ledger evidence; promotion uses the sole typed authorization event; activation requires an explicit evidence checklist, actor and audit event. |
| Blocking/product/compliance decision | D-PAY-01 - Select processor, card/ACH capabilities, webhook/security/settlement semantics and operational ownership. | Luke | P08-02, P08-03, P08-04; Before P08-02 adapter | D-PAY-01 - Choose only after domain/ledger contract; retain processor-neutral ports. |
| Blocking/product/compliance decision | D-COM-05 - Select financing partner and define when funding is sufficiently authorized/settled; otherwise hide the path. | Luke | P07-02, P08-02; Before financed option is exposed | D-COM-05 - Do not block card/ACH MVP; keep optional and disabled. |
| Blocking/product/compliance decision | D-COM-06 - Decide retained revenue versus future-journey credit after heartbeat loss and contractual support window. | Luke | P08-03; Before automated pregnancy-stage refund logic | D-COM-06 - Manual/counsel-reviewed hook until policy is approved; never hard-code. |

| Type | ID / risk | Owner | Affected PRs / latest point | Treatment |
|---|---|---|---|---|
| Blocking/product/compliance decision | D-COM-07 - Ratify earned/uncharged/refunded treatment, GC withdrawal/replacement and pay-in-full/financed reconciliation. | Luke | P08-03; Before real billing | D-COM-07 - Use document schedule as test anchor; require legal/finance approval for live policy. |
| Blocking/product/compliance decision | D-COM-08 - Define reminder cadence, grace period, pauseable services and safety-critical/document access that can never be blocked. | Luke | P08-04; Before P08-04 | D-COM-08 - Default to continued safety and entitled-document access; pause only clearly non-critical future services. |
| Implementation risk | Double charge/refund | Luke or named PR DRI | Package PRs; before associated gate | Mitigate through the explicit tests, rollout and forward-repair plan |
| Implementation risk | Processor lock-in | Luke or named PR DRI | Package PRs; before associated gate | Mitigate through the explicit tests, rollout and forward-repair plan |
| Implementation risk | Clinical timing harm | Luke or named PR DRI | Package PRs; before associated gate | Mitigate through the explicit tests, rollout and forward-repair plan |
| Implementation risk | Pass-through commingling | Luke or named PR DRI | Package PRs; before associated gate | Mitigate through the explicit tests, rollout and forward-repair plan |
| Implementation risk | Policy gaps | Luke or named PR DRI | Package PRs; before associated gate | Mitigate through the explicit tests, rollout and forward-repair plan |

## 1.8 Existing-work salvage

| Source | Head | Valuable content | Destination | Recommendation | Extraction status/safety |
|---|---|---|---|---|---|
| No direct source | - | Build from approved contracts/evidence | Package PRs | New implementation | N/A |

Package source inventory:

- Business-model schedules and smoke paths; no current code is authoritative

## 1.9 Ownership recommendation

Luke leads this work package, with Alan providing technical review. The split is proposed for review and can be amended before implementation begins.

| PR | DRI | Implementer | Reviewer | Final approver | External/high-capability review |
|---|---|---|---|---|---|
| P08-01 | Luke | Luke (proposed) | Alan | Joint | Mandatory |
| P08-02 | Luke | Luke (proposed) | Alan | Joint | Mandatory |
| P08-03 | Luke | Luke (proposed) | Alan | Joint | Mandatory |
| P08-04 | Luke | Luke (proposed) | Alan | Joint | Mandatory |

Luke leads implementation and product acceptance for this package; Alan reviews architecture, migration, security and failure behavior. This is a recommendation pending joint review, not a compulsory assignment.

## 1.10 Definition of done

- ☐ Every proposed PR is merged in dependency order or explicitly dispositioned, with its objective acceptance criteria and review record complete.
- ☐ All package blocking decisions are resolved by the named owner; no commercial/privacy/security value is guessed in code.
- ☐ Local, PostgreSQL, role/authorization, retry/concurrency, failure, mobile and synthetic-cloud evidence required by this package is attached.
- ☐ Cross-repository contracts and artifact/migration versions are pinned; partial-merge behavior is fail-closed or explicitly degraded.
- ☐ Observability is sufficient to identify revision, transition, dependency and safe failure without sensitive payloads.
- ☐ Existing PR/worktree value and reviewer findings are extracted before closure; no cleanup occurs without team confirmation.
- ☐ The associated programme gate is accepted by Luke and any required external reviewer; tests/deploy alone do not establish B1/B2.